A Unified SAM-Guided Self-Prompt Learning Framework for Infrared Small Target Detection

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***Abstract*—Infrared small target detection (ISTD) aims to precisely capture the location and morphology of small targets under all-weather conditions. Compared with generic objects, infrared targets in remote fields of view are smaller in size and exhibit lower signal-to-clutter ratios (SCRs). This poses a significant challenge in simultaneously preserving low-level target details and understanding high-level contextual semantics, forcing a tradeoff between reducing miss detection and suppressing false alarms. In addition, most existing ISTD methods are designed for specific target types under certain infrared platforms, rather than as a unified framework broadly applicable across diverse infrared sensing scenarios. To address these challenges, we propose a unified self-prompt learning (SPL) framework for ISTD under the guidance of the segment anything model (SAM). Specifically, the model is incorporated with the SAM in the encoding stage through a consult-guide manner, adapting the general knowledge to facilitate task-specific contextual understanding. Then, shallow-layer features are employed to generate self-derived prompts, which bidirectionally interact with encoded latent representations to complement subtle low-level details. Moreover, the semantic inconsistency during resolution recovery is mitigated by integrating a mutual calibration module into skip connections, ensuring coherent spatial–semantic fusion. Extensive experiments are conducted on four public ISTD datasets, and the results demonstrate that the proposed method consistently achieves superior performance across different infrared sensing platforms and target types. The code is released at https://github.com/fuyimin96/SAM-SPL**

***Index Terms*—Infrared small target detection (ISTD), remote sensing, segment anything model (SAM), self-prompt learning (SPL).**

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I. Introduction

INFRARED small target detection (ISTD) [1] plays a crucial role in various military and civilian applications, such as maritime surveillance and airspace monitoring. Unlike common optical devices [2], [3], [4], infrared sensing [5], [6] relies on the passive reception of thermal radiation, thereby enabling all-weather perception of target location and morphology. However, as shown in Fig. 1, specific characteristics of infrared small targets also introduce additional challenges compared to general object detection [7]. First, targets captured from long distances appear extremely small in size, resulting in a lack of discriminative features. Second, targets in infrared images typically exhibit low signal-to-clutter ratios (SCRs), making them easily submerged within complex backgrounds. These challenges severely increase false alarm and miss detection risks, necessitating both precise detail capture and holistic contextual understanding.

Traditional ISTD methods rely on handcrafted features and predefined mathematical models to distinguish targets from background and can be categorized into filter-based [8], [9], [10], local contrast-based [11], [12], [13], [14], and low-rank-based methods [15], [16], [17]. While interpretable and efficient, the model-driven methods suffer from poor generalization across dynamic scenes, failing to meet practical requirements. In comparison, data-driven methods have achieved continuous performance improvements, benefiting from the powerful representation abilities of deep neural networks (DNNs) [18], [19], [20]. Existing data-driven methods typically employ an encoder–decoder architecture [21], [22], [23], [24] to jointly learn target details and contextual semantics. Nevertheless, these two objectives are inherently difficult to reconcile, as deepening the network causes the loss of local target details, whereas shallow architectures fail to capture sufficient contextual semantics. Although some methods fuse multiscale features to complement low-level details, the lack of interaction with semantic representations severely limits the restoration of spatial structures. Besides, when adopting skip connections to integrate low-level details into the decoding stage, the semantic inconsistency between encoder and decoder features also impairs target awareness throughout the resolution recovery process. Consequently, existing data-driven methods struggle to accurately detect targets from complex backgrounds while effectively suppressing false alarms [25], raising serious reliability concerns. In addition, the learning frameworks of most existing methods are tailored to specific target types, rather than adhering to a unified paradigm. As a result, the acquired task-specific knowledge exhibits limited applicability across different infrared sensing platforms (e.g., aerial or satellite), severely restricting its adaptability in real-world deployments.

Empirically, the understanding of contextual semantics correlates strongly with the general cognitive capability, whereas the capture of target details relies more on task-specific learning. Building upon this, an intuitive solution to break the semantic-detail bottleneck is to leverage the powerful general knowledge encoded in large foundation models [26] as semantic guidance. Recently, the segment anything model (SAM) [27], [28] has emerged as a versatile foundation model for visual perception, achieving remarkable performance across diverse tasks [29], [30], [31]. However, as illustrated in Fig. 2, even with explicit prompts, the SAM fails to accurately capture the location or morphology of targets when directly applied to ISTD. This failure primarily arises from the disparity between the visual characteristics of infrared small targets and the generic objects used to train SAM. Therefore, the effective adaptation of general knowledge offers a more sensible solution for facilitating contextual understanding in ISTD, rather than directly employing the entire SAM architecture.

To address the aforementioned challenges, we propose a unified SAM-guided self-prompt learning (SAM-SPL) framework for ISTD. Unlike existing data-driven ISTD methods, SAM-SPL adapts the general knowledge of foundation models to provide semantic guidance for contextual understanding, rather than learning entirely from scratch. First, the SAM encoder inference is incorporated with feature extraction blocks in a consult-guide manner to facilitate the understanding of contextual semantics. Then, shallow-layer features are employed to generate self-derived prompts, which complement low-level details in deep semantic space by establishing bidirectional interactions with encoded latent representations. Subsequently, a mutual calibration module is integrated into skip connections to ensure coherent feature fusion, effectively enhancing target awareness by mitigating semantic inconsistency. Finally, the overall framework is optimized by a multiscale supervision strategy to achieve robust perception across varying sizes and shapes. We conduct extensive experiments on four ISTD datasets, and the results demonstrate that SAM-SPL consistently achieves superior detection performance for various target types across different infrared sensing platforms.

Our main contributions can be summarized as follows.

1) We propose SAM-SPL, a unified ISTD framework with broad applicability across diverse sensing scenarios. Different from conventional paradigms, SAM-SPL breaks the semantic-detail bottleneck by leveraging the general knowledge of the SAM through consult-guide adaptation.
2) We propose a self-prompt learning (SPL) strategy to complement low-level details in deep semantic space for improved spatial structure restoration. The long-overlooked lack of semantic-detail collaboration is effectively addressed by establishing bidirectional interactions with encoded representations.

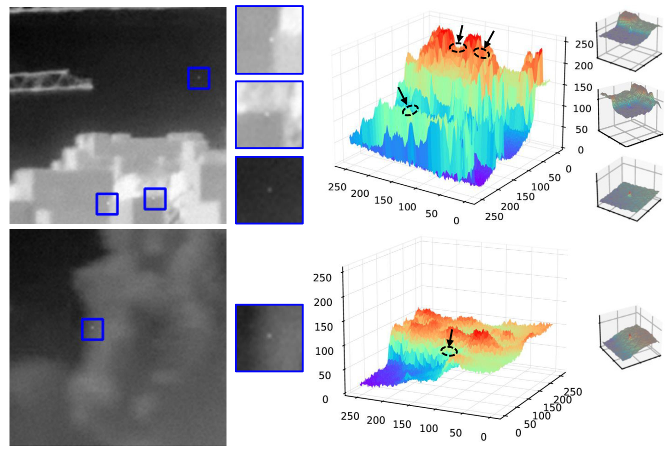


Fig. 1. Infrared images and zoomed-in target regions with corresponding 3-D surface visualizations. The targets are imperceptible and easily submerged in background clutter.

3) We develop a mutual calibration module that refines skip connections by mitigating semantic inconsistency between cross-stage representations. This ensures coherent spatial–semantic fusion during resolution recovery for better target awareness.

The remainder of this article is organized as follows. Section II briefly reviews the current research related to our work. Section III introduces the technical details of our proposed method. Section IV presents the experimental results and corresponding analyses. Finally, Section V concludes the work and outlines future research directions.

II. Related Works

In this section, we first review current progress in ISTD. After that, we introduce recent advances in the SAM and prompt learning, which are correlated to our methodological design.

A. Infrared Small Target Detection

From the perspective of input data, the task setup of ISTD can be divided into single-frame and multiframe scenarios. This work focuses on the single-frame ISTD, which is more suitable for real-world deployment due to high computational efficiency. Notably, in the context of ISTD, the detection typically refers to segmenting the targets from the background rather than predicting bounding boxes, as the latter fails to capture target morphology.

Before the widespread adoption of DNNs, traditional model-driven methods aimed to separate targets from backgrounds by modeling their statistical differences. Filtering-based methods [8], [9] design specific filters to suppress background clutter according to disparities between targets and the background in the spatial or frequency domain. Inspired by the human visual system, local contrast-based methods [11], [12], [13], [14] identify targets by measuring intensity contrast between the central patch and its surroundings. Based on structural characteristics, low-rank-based methods [15], [16], [17], [32], [33] extract targets as sparse components from the background

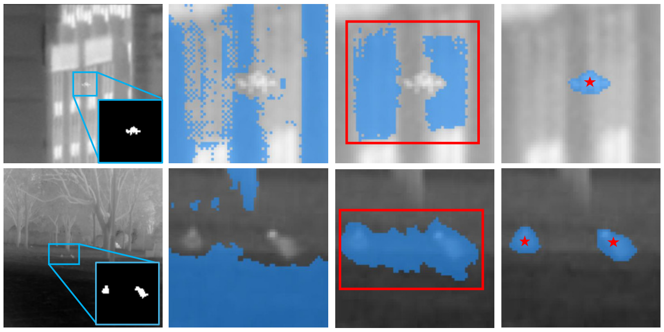


Fig. 2. Visualizations of directly applying the SAM to ISTD under prompt-free, box-prompt, and point-prompt settings, respectively.

through image decomposition and reconstruction. While interpretable and computationally efficient, the applicability of model-driven methods to dynamic task scenarios is severely limited due to the reliance on prior assumptions and fixed hyperparameters.

Driven by the superior representation capacity of DNNs and the availability of public datasets, data-driven methods have become the dominant paradigm for ISTD. Dai et al. [34] enriched the cross-layer information fusion through asymmetric contextual modulation (ACM) to improve the perception of infrared small targets. To better retain target details, Dai et al. [35] proposed an attentional local contrast network (ALCNet) by introducing a physically interpretable local contrast measurement. Zhang et al. [36] designed a novel infrared shape network (ISNet), which leverages shape reconstruction for precise morphology prediction. In addition, attention mechanisms have been widely adopted to strengthen target-relevant representations. Aiming at preserving small targets in deep layers, Li et al. [37] proposed a dense nested attention network (DNA-Net) to realize hierarchical feature interaction and enhancement. Zhang et al. [38] integrated an attention-guided context block into the pyramid architecture to highlight small targets against backgrounds. Sun et al. [39] proposed a multidirection-guided attention mechanism to enhance the saliency of infrared targets. Zhang et al. [38] effectively suppressed background noise interference by leveraging feature compensation and correlation across different levels. Meanwhile, some works also attempt to develop ISTD-oriented network architectures and learning objectives. Wu et al. [40] designed a nested U-Net-in-U-Net (UIU-Net) architecture to avoid the loss of local contrast information during downsampling. Liu et al. [41] proposed a scale- and location-aware loss to improve the sensitivity of a multiscale head network (MSHNet) to small targets. To better align with the spatial characteristics of infrared small targets, Yang et al. [42] designed a novel pinwheel-shaped convolution to enhance low-level feature extraction. Wu et al. [43] unfolded the robust principal component analysis optimization into the network (RPCANet) for interpretable detection. Taking domain knowledge into account, Wu et al. [44] inserted a learnable local saliency kernel module (L2SKM) into DNNs to ensure discriminative feature extraction of size-varying targets. Zhang et al. [45] designed a Runge–Kutta transformer to enhance semantic discriminability and employed a wavelet-structured regularized pruning approach [46] to boost detection efficiency. Apart from aerial- and land-based platforms, Wu et al. [47] proposed a multilevel TransUNet [48] (MTU-Net) for satellite-based infrared ship detection, which is more challenging due to increased imaging distance and background complexity. Nevertheless, existing data-driven ISTD methods largely depend on the model to jointly learn target details and contextual semantics from scratch, which is hindered by the inherent conflict between these two objectives.

B. Segment Anything Model

Pretrained on a large-scale corpus of natural images, the SAM [27] has emerged as a prompt-driven vision foundation model with strong zero-shot generalization across generic segmentation tasks. Furthermore, SAM2 [28] extends the SAM to the video domain by incorporating a streaming memory module to realize spatiotemporal understanding.

Benefiting from its rich general knowledge, the effectiveness of the SAM has also been adapted to various applications with advanced performance. Mazurowski et al. [29] simulated interactive prompts to apply the SAM for medical image analysis across diverse modalities and anatomies. Song et al. [30] leveraged the SAM to enhance the robustness of multimodal 3-D object detection against noise scenarios by adaptive fusion. Lin et al. [31] proposed a SAM-based two-stage framework for 6-D pose estimation through geometry-aware point matching. To match objects across video frames in complex scenes, Li et al. [49] employed SAM predications as dense region proposals to model instance-level correspondence. Zuo et al. [50] refined the scene understanding results by aligning the predictions with SAM segments through a voting strategy. Beyond standard computer vision domains, the adaptation of the SAM has also achieved notable success in the field of remote sensing. Wang et al. [51] utilized detection annotations to exploit the potential of the SAM in generating precise pixel-level masks for remote sensing images, effectively reducing manual labeling burden. Feng et al. [52] used explicit visual prompting to fine-tune a SAM variant for road extraction by a frequency adapter strategy. To mitigate reliance on large-scale training data, Ding et al. [53] introduced a convolutional adaptor into a lightweight SAM architecture to extract semantic latent for change detection. Shan et al. [54] fine-tuned the SAM via low-rank adaptation [55] to bridge the domain gap between natural and remote sensing imagery, achieving accurate interactive segmentation of moving objects. Some recent studies have explored adapting the SAM to camouflaged object detection [56], [57], [58], a representative challenging scenario where targets are visually inconspicuous due to seamless blending into the background. However, the adaptation of SAM visual priors to ISTD remains severely constrained by the domain gap between infrared and natural images, along with the extreme scale disparity between small and generic objects. Despite the recent emergence of IRSAM [59], a customized SAM architecture for ISTD, its focus remains on structural redesign rather than the efficient adaptation of general knowledge.

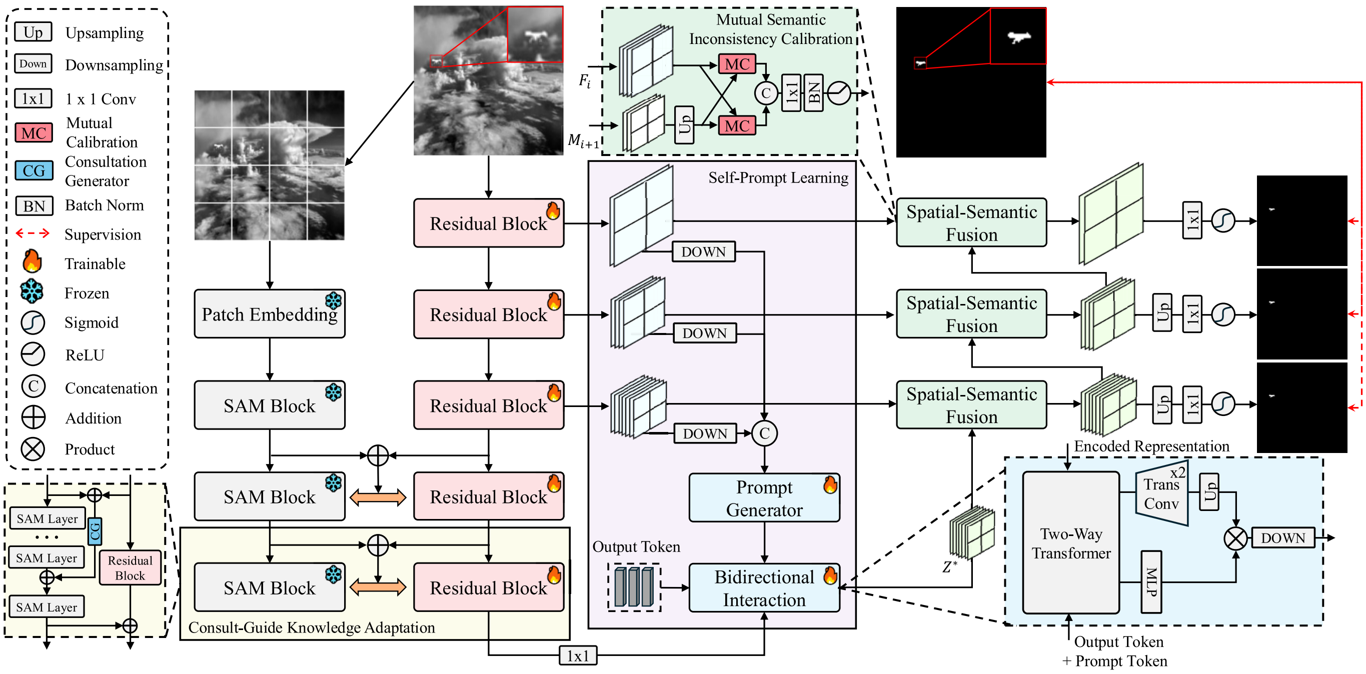


Fig. 3. Overall pipeline of our proposed SAM-SPL. Built upon an encoder–decoder architecture, SAM-SPL consists of three main components: CGKA, SPL, and MSIC. Predictions from all decoder layers are used to optimize the model through a multiscale supervision strategy.

C. Visual Prompt Learning

Prompt learning [60], [61] initially emerged in natural language processing (NLP) tasks to repurpose pretrained language models for specific downstream tasks. Encouraged by its superiority in modeling task-specific context, prompt learning techniques have been widely adopted in computer vision tasks. As an early attempt in visual prompt learning, Jia et al. [62] introduced visual prompt tuning as a lightweight alternative to full fine-tuning of vision models. Wang et al. [63] optimized learnable prompts in the memory space as instructions for continuous learning of pretrained models. Moving beyond single-modal image data, Khattak et al. [64] designed a multimodal prompting strategy to improve vision–language alignment. Wang et al. [65] proposed a transferable prompt learning paradigm that enables simultaneous adaptation to multiple downstream tasks. Zhang et al. [66] employed a dynamic template prompter to model temporal appearance variations, enabling efficient and accurate object tracking. To overcome diverse types of degradation, Potlapalli et al. [67] proposed an all-in-one image restoration framework that leverages input-conditioned prompts to guide the reconstruction process. With respect to remote sensing, Zhao et al. [68] generated diverse text prompts to improve the classification accuracy of remote sensing images with limited samples. Yao et al. [69] proposed a hierarchical mask prompting module to suppress background interference for object detection in aerial remote sensing images. Kong et al. [70] generalized vision–language models to multisource remote sensing imagery classification by learning spectral prompts.

III. Methodology

In this section, we first present the overall pipeline of our proposed method for capturing the location and morphology of small targets in infrared images. Then, we provide the technical details of each main component, which are, respectively, designed to enhance contextual understanding, supplement low-level details, and mitigate semantic inconsistency.

A. Pipeline Overview

The overall pipeline of SAM-SPL is illustrated in Fig. 3. Given an input image $x \in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote the height, width, and channel, respectively. Multilevel feature maps $F = [F_0, \ldots, F_4]$ are extracted through five stacked convolutional residual blocks. At each level $l \in \{0, \ldots, 4\}$, the spatial resolution of feature maps F_l is progressively downsampled to extract semantic abstraction, that is, $F_l \in \mathbb{R}^{H/2^l\times W/2^l\times C_l}$, where $C_l = 24 \cdot 2^l$ for each layer. In parallel, the image is transformed into patch embeddings and propagated through the frozen SAM2 encoder, whose blocks are sequentially incorporated with the corresponding convolutional blocks to perform knowledge adaptation. Then, shallow features from early convolutional blocks of the encoder are used to generate self-derived prompts, which are bidirectionally interacted with latent representations to supplement low-level details. Afterward, the decoded features are gradually recovered to the initial input resolution, with mutual calibration integrated into the skip connection. During training, outputs from all decoder stages are converted into prediction maps to perform multiscale supervision. For inference, only the final-stage predictions are used to ensure computational efficiency.

B. Consult-Guide Knowledge Adaptation

Existing data-driven ISTD methods typically follow a routine learning pipeline that acquires task-specific knowledge by learning from scratch. In this case, the understanding of contextual semantics often heavily relies on increasing network depth. However, blindly deepening the network does not guarantee better contextual understanding, but rather undermines the preservation of target details. Since contextual understanding is closely tied to the model's general cognitive capacity, principled adaptation of the powerful general knowledge encoded in SAM offers a more effective means of facilitating task-specific contextual understanding.

Motivated by this, we employ the SAM encoder to provide semantic guidance through a consult-guide knowledge adaptation (CGKA) strategy, where multilevel cognitions are interacted through progressive knowledge acquisition and consultation. Specifically, the first three stages of the SAM2 encoder (with strides of 4, 8, and 16) are aligned with the last three convolutional residual blocks, constructing a hierarchical symmetric structure with consistent spatial resolution. At each hierarchy $h \in \{0, 1, 2\}$, feature representations S_h from the SAM2 encoder and the corresponding convolutional residual block are integrated into E_h

$$E_h = S_h \oplus F_l \tag{1}$$

where $l = h + 2$ and $\oplus$ denotes element-wise addition operation. Then, the integrated representations are passed through a consultation generator (CG) to formulate semantic enquiries

$$I_h = \mathrm{CG}(E_h) = \mathrm{Conv}_{7\times 7}(\mathrm{Conv}_{3\times 3}(E_h)) \tag{2}$$

where each convolutional layer is followed by batch normalization and GELU activation. Subsequently, the generated enquiries I_h are injected into the next SAM stage, from which the generated semantic guidance is routed to the encoder for adapting general knowledge to ISTD

$$\hat{F}_{l+1} = F_{l+1} \oplus (\mathrm{SAM}(I_h \oplus \bar{S}_{h+1})) \tag{3}$$

where $\bar{S}_{h+1}$ denotes the intermediate representation at the $(h + 1)$th stage before the final SAM layer. Through such enquiry-driven semantic guidance, the domain gap between pretraining corpora of the SAM and infrared images can be effectively bridged. Thereafter, the adapted representations and associated SAM counterparts are used to generate subsequent semantic enquiries and drive the knowledge adaptation at the next hierarchy. After completing the entire adaptation process, a comprehensive semantic understanding of infrared small targets is embedded into the resulting encoded representation $Z \in \mathbb{R}^{H/2^4 \times W/2^4 \times 384}$ (i.e., $\hat{F}_4$).

C. Self-Prompt Learning

In addition to holistic contextual understanding, the effectiveness of ISTD also hinges on the precise capture of target details. However, repeated downsampling during semantic extraction inevitably results in the loss of target details in deep semantic spaces. Although some solutions incorporate multiscale features for compensation, the restoration of spatial structures remains suboptimal due to the lack of collaborative interaction with semantic representations.

To overcome this limitation, we generate self-derived prompts from shallow-layer features, which complement low-level details through bidirectional interaction with deep semantic representations. First, the shallow-layer feature maps $[F_0, F_1, F_2]$ are downsampled to the same spatial resolution as Z and concatenated along the channel dimension

$$F_s = \mathrm{Concat}[\mathcal{D}_0(F_0), \mathcal{D}_1(F_1), \mathcal{D}_2(F_2)] \tag{4}$$

where $\mathcal{D}_i(\cdot)$ $(i = 0, 1, 2)$ correspond to downsampling operations with strides of 16, 8, and 4, respectively. Next, the prompt tokens $P \in \mathbb{R}^{H/2^4 \times W/2^4 \times d}$ are derived from F_s through a prompt generator (PG)

$$P = \mathrm{PG}(F_s) = \mathrm{Conv}_{1\times 1}(F_s) \tag{5}$$

where the channel dimension of P matches the embedding dimension of the SAM mask decoder (i.e., $d = 256$). Then, a learnable output token $O \in \mathbb{R}^d$ is appended to the prompt tokens

$$T = \mathrm{Concat}[P, O]. \tag{6}$$

Afterward, the combined tokens interact with the encoded representation via a two-way transformer to realize bidirectional information exchange. In this process, T and Z alternately serve as query and key–value pairs for cross-attention. To ensure compatibility, the channel dimension of Z is aligned with that of T via a 1×1 convolution

$$\begin{aligned} T' &= \mathrm{MLP}(\mathrm{CA}(\mathrm{SA}(T), Z, Z)) \\ Z' &= \mathrm{CA}(Z, T', T') \end{aligned} \tag{7}$$

where $\mathrm{SA}(\cdot)$, $\mathrm{CA}(\cdot)$, and $\mathrm{MLP}(\cdot)$ represent the self-attention, cross-attention, and multilayer perceptron modules, respectively. After completing iterative state updates, the representation Z' undergoes two transposed convolution layers to transfer the spatial resolution consistent with F_3

$$\tilde{Z} = \mathrm{TransConv}_{2\times 2}(\mathrm{TransConv}_{2\times 2}(Z')) \tag{8}$$

while the channel dimension of $\tilde{Z}$ is identical to C_2. Meanwhile, the output token is projected to the same channel dimension through a three-layer MLP. Finally, we employ the output token to refine the latent representations that fulfill the requirements of both contextual understanding and detail capture for ISTD

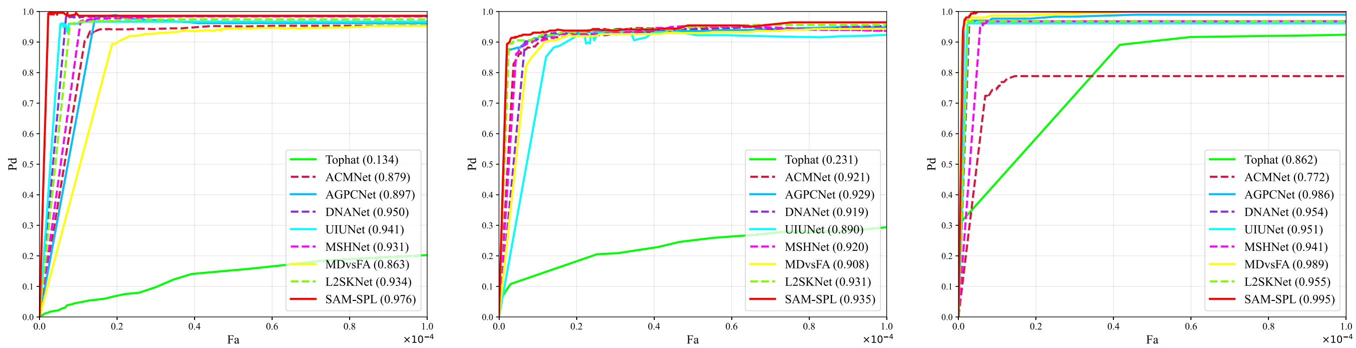
$$Z^* = \mathcal{D}(\tilde{Z} \otimes \mathrm{MLP}(T'[-1])) \tag{9}$$

where $\mathcal{D}(\cdot)$ denotes a downsampling operation with stride 2 and $\otimes$ is the point-wise product operation.

D. Mutual Semantic Inconsistency Calibration

Upon obtaining the enhanced contextual understanding and enriched target details, effective resolution recovery and supervision strategies are critical for accurate target perception. Existing methods often adopt skip connections between multilevel encoder and decoder features during resolution recovery. However, the semantic inconsistency between cross-stage representations can impair the formation of target awareness.

To mitigate this impairment, we integrate mutual calibration modules into the skip connections to enhance target perception. During the standard resolution recovery process, at each decoding stage, the latent representation is progressively fused



(a) (b) (c)

Fig. 4. ROC curves of SAM-SPL and some representative ISTD methods on the (a) NUDT-SIRST, (b) IRSTD-1K, and (c) IRSTDID-SKY datasets. The AUC values are reported in parentheses next to each method in the legend.

TABLE I

Comparison of ISTD Performance on the NUDT-SIRST, IRSTD-1K, and IRSTDID-SKY Datasets. The Best and the Second-Best Results Are Colored in Red and Cyan, Respectively

Method	Venue	NUDT-SIRST				IRSTD-1K				IRSTDID-SKY			
		$IoU(\%)$	$F_1(\%)$	$P_d(\%)$	$F_a(10^{-6})$	$IoU(\%)$	$F_1(\%)$	$P_d(\%)$	$F_a(10^{-6})$	$IoU(\%)$	$F_1(\%)$	$P_d(\%)$	$F_a(10^{-6})$
Tophat [8]	OE_{96}	22.26	36.38	91.32	771.20	5.59	10.60	69.36	702.30	3.39	6.56	19.87	1.13
Max-Median [9]	SPIE_{99}	4.20	7.64	58.41	36.89	7.00	8.15	65.21	59.73	8.31	15.34	76.28	68.02
LIG [10]	IPT_{18}	38.63	55.74	82.85	133.72	19.30	32.25	72.05	175.17	39.54	56.67	85.89	43.80
MPCM [11]	PR_{16}	9.26	16.95	67.30	316.60	12.54	22.30	70.03	44.24	17.44	29.70	93.52	223.67
RLCM [12]	GRSL_{18}	18.54	31.27	59.68	64.89	22.36	36.55	72.72	73.20	18.70	31.51	51.92	15.18
TLLCM [13]	GRSL_{19}	27.85	43.57	61.53	19.39	15.39	26.67	76.76	110.95	27.85	43.57	61.53	19.39
WSLCM [14]	GRSL_{20}	10.27	18.62	88.46	1812.65	6.80	12.73	88.21	1097.06	15.08	26.20	96.15	323.17
MSLSTIPT [15]	TGRS_{20}	9.47	17.29	59.89	104.26	17.82	30.23	61.61	91.32	16.33	28.08	75.64	7.55
IPI [16]	TIP_{13}	34.62	51.43	85.71	106.30	17.87	30.33	71.38	93.17	49.85	66.54	91.03	20.77
PSTNN [33]	RS_{19}	25.46	40.58	71.64	97.04	16.36	28.12	62.96	77.87	54.71	31.48	93.58	59.4
MDvsFA [25]	ICCV_{19}	73.34	84.62	93.54	41.04	62.16	76.67	91.92	22.93	69.78	82.20	96.79	3.23
ACMNet [34]	WACV_{21}	69.98	82.33	95.77	17.90	62.92	77.21	90.57	17.31	30.91	47.23	74.36	8.64
ALCNet [35]	TGRS_{21}	85.03	91.91	97.89	18.20	63.52	77.61	85.86	12.96	41.74	58.90	87.17	5.76
ISNet [36]	CVPR_{22}	87.77	93.37	95.56	9.42	64.73	77.99	87.21	15.03	69.06	81.70	96.15	2.17
DNANet [37]	TIP_{22}	93.20	96.48	98.20	9.38	65.44	79.05	88.55	9.43	70.71	82.84	96.79	2.92
AGPCNet [38]	TAES_{23}	85.35	92.09	97.35	6.78	64.92	78.72	89.56	18.56	65.38	79.07	97.04	2.08
UIUNet [40]	TIP_{22}	92.13	95.89	95.24	1.72	65.37	79.01	89.23	21.52	69.76	82.19	96.15	2.79
MTUNet [47]	TGRS_{23}	84.02	91.32	97.99	7.49	65.24	78.96	89.56	9.51	71.05	83.07	96.79	2.44
MSHNet [41]	CVPR_{24}	76.49	86.67	96.08	26.40	65.50	77.52	91.58	44.62	63.58	77.73	95.51	6.38
RPCANet [43]	CVPR_{24}	90.88	95.22	96.72	9.33	62.73	77.08	87.88	15.56	67.46	80.57	96.79	1.99
L2SKNet [44]	TGRS_{24}	93.83	96.81	98.31	2.60	65.75	78.85	91.25	16.38	71.21	83.19	96.79	2.97
P-MSHNet [42]	AAAI_{25}	78.59	88.01	96.82	13.47	59.94	74.95	91.83	15.03	63.86	77.95	96.79	4.67
SAM-SPL		94.63	97.24	99.47	2.55	74.09	85.11	92.59	9.28	73.40	84.66	98.72	0.97

with the feature map F_i from the corresponding encoder stage ($i = 0, 1, 2$) to produce the upsampled feature map M_i

$$M_i = \begin{cases} \mathcal{G}_i\left(\mathrm{Concat}[\mathrm{Up}\left(Z^*\right), F_i]\right), & \text{if } i = 2 \\ \mathcal{G}_i\left(\mathrm{Concat}[\mathrm{Up}\left(M_{i+1}\right), F_i]\right), & \text{otherwise} \end{cases} \tag{10}$$

where each fusion block $\mathcal{G}_i$ consists of a 1×1 convolution, batch normalization, and a ReLU activation function. In SAM-SPL, before the concatenation of cross-stage representations, encoder and decoder features mutually serve as calibration cues to mitigate semantic inconsistency

$$\bar{F}_i = F_i + W_i^F(\mathrm{GAP}(M_{i+1}))$$

$$\bar{M}_{i+1} = \mathrm{Up}(M_{i+1}) + W_i^M(\mathrm{GAP}(F_i)) \tag{11}$$

where $\mathrm{GAP}(\cdot)$ denotes global average pooling and W_i^F and W_i^M are learnable linear projections. Benefitting from the coherent spatial–semantic fusion, the target awareness is effectively enhanced. Moreover, considering the varying sizes and shapes of infrared small targets, we use a multiscale supervision strategy during training

$$L = -\sum_{i=0}^{2}\left(Y\log\left(\hat{Y}_i\right) + (1 - Y)\log\left(1 - \hat{Y}_i\right)\right) \tag{12}$$

TABLE II
COMPARISON OF ISTD PERFORMANCE ON THE NUDT-SIRST-SEA DATASET. THE BEST AND THE SECOND-BEST RESULTS OF DATA-DRIVEN METHODS ARE COLORED IN RED AND CYAN, RESPECTIVELY

Model-Driven Methods	Venue	$IoU(\%)$	$P_d(\%)$	$F_a(10^{-6})$	Data-Driven Methods	Venue	$IoU(\%)$	$P_d(\%)$	$F_a(10^{-6})$
Tophat [8]	OE_{96}	1.17	10.39	95.37	ACMNet [34]	$WACV_{21}$	47.57	70.46	21.31
Max-Median [9]	$SPIE_{99}$	0.28	6.62	46.17	ALCNet [35]	$TGRS_{21}$	48.90	58.65	9.13
LIG [10]	IPT_{18}	2.48	17.99	269.86	ResUNet [24]	$JPRS_{20}$	41.05	60.18	17.19
MPCM [11]	PR_{16}	3.75	17.13	102.26	MDvsFA [25]	$ICCV_{19}$	0.37	0.18	92.00
RLCM [12]	$GRSL_{18}$	3.92	35.13	173.64	DNANet [37]	TIP_{22}	42.17	61.60	17.19
TLLCM [13]	$GRSL_{19}$	0.59	10.27	14.41	UIUNet [40]	TIP_{22}	44.13	66.61	15.70
WSLCM [14]	$GRSL_{20}$	0.60	8.02	7.33	MTUNet [47]	$TGRS_{23}$	59.98	81.22	16.64
MSLSTIPT [15]	$TGRS_{20}$	0.33	6.68	6283.00	MSHNet [41]	$CVPR_{24}$	47.34	74.58	16.27
IPI [16]	TIP_{13}	4.79	24.78	288.40	RPCANet [43]	$CVPR_{24}$	2.79	53.38	892.68
RIPT [17]	$JSTARS_{17}$	0.36	6.32	1.92	L2SKNet [44]	$TGRS_{24}$	40.71	53.66	15.53
NRAM [32]	RS_{18}	0.39	5.89	3.59	P-MSHNet [42]	$AAAI_{25}$	46.68	64.82	15.03
PSTNN [33]	RS_{19}	1.50	7.59	15.44	SAM-SPL		60.93	82.33	14.92

where $\hat{Y}_i \in \mathbb{R}^{H\times W}$ is the prediction map of each stage

$$\hat{Y}_i = \sigma\left(\mathrm{Conv}_{1\times 1}\left(\mathrm{Up}\left(M_i\right)\right)\right) \tag{13}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function, and the upsampling operation $\mathrm{Up}(\cdot)$ is implemented through bilinear interpolation. At inference time, the final-stage prediction map is converted to a binary segmentation map as the detection result

$$\hat{Y}(i,j) = \begin{cases} 1, & \text{if } \hat{Y}_0(i,j) \geq \delta \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where the threshold δ is set to 0.5 and i and j are spatial indices over the height and width.

IV. Experiments

A. Experimental Setup

1) Dataset: We thoroughly evaluate the proposed method across four public ISTD datasets, encompassing diverse target types captured by aerial-, land-, and satellite-based platforms.

1) *NUDT-SIRST* [37] contains 1327 synthesized infrared images of size 256 × 256. Various types of targets with different poses are introduced across diverse background scenes to ensure physical plausibility and visual realism.
2) *IRSTD-1K* [36] provides 1000 realistic infrared images sized 512 × 512, where targets in various shapes and sizes are accurately annotated at the pixel level.
3) *IRSTDID-SKY* [71] consists of 500 scene-rich infrared images with small targets under real stripe noise perturbations, which closely aligns with practical task scenarios. All images are resized into 480 × 480.
4) *NUDT-SIRST-Sea* [47] comprises large satellite-based infrared images captured under long-range imaging conditions for tiny ship detection. Each image covers about 10 000 km^2 sea area and is cut into 1024 × 1024 patches.

2) Evaluation Metrics: To prove the effectiveness of the proposed method, the following metrics are used to report the ISTD performance at both pixel and target levels.

1) *Intersection over union (IoU)* measures the pixel-level intersection between predicted and ground-truth masks to indicate localization accuracy

$$\mathrm{IoU} = \frac{A_{\mathrm{inter}}}{A_{\mathrm{union}}} = \frac{\sum_{i=1}^{N} TP[i]}{\sum_{i=1}^{N}(T[i] + P[i] - TP[i])} \tag{15}$$

where A_{inter} and A_{union} refer to areas of interaction and union regions across N samples, respectively. T, P, and TP represent the number of ground truth, predicted, and true positive pixels, respectively.
2) *F-measure (F_1)* reflects the tradeoff between miss detection and false alarm at the pixel level

$$F_1 = \frac{2 \times \mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{16}$$

3) *Probability of detection (P_d)* quantifies the ratio of correctly detected targets N_{detect} to total targets N_{total}

$$P_d = \frac{N_{\mathrm{detect}}}{N_{\mathrm{total}}}. \tag{17}$$

A target is considered to be correctly detected if the deviation between its predicted and ground-truth centroid is less than 3 [37].
4) *False alarm rate (F_a)* quantifies the proportion of false predicted pixels P_{false} relative to the total pixels P_{total}

$$F_a = \frac{P_{\mathrm{false}}}{P_{\mathrm{total}}} \tag{18}$$

where pixels of predicted targets with centroid deviations over 3 are regarded as false predictions.
5) *Receiver operation characteristics (ROC)* curves provide a comprehensive evaluation of P_d against F_a across varying thresholds, where a larger area under the curve (AUC) indicates superior overall performance.

3) General Implementation: In our experiments, we follow the training–testing split strategies used in previous works for all datasets [36], [37], [47], except for IRSTDID-SKY, where a unified 4:1 ratio is applied due to the absence of an official split. We adopt a U-Net encoder–decoder architecture with five

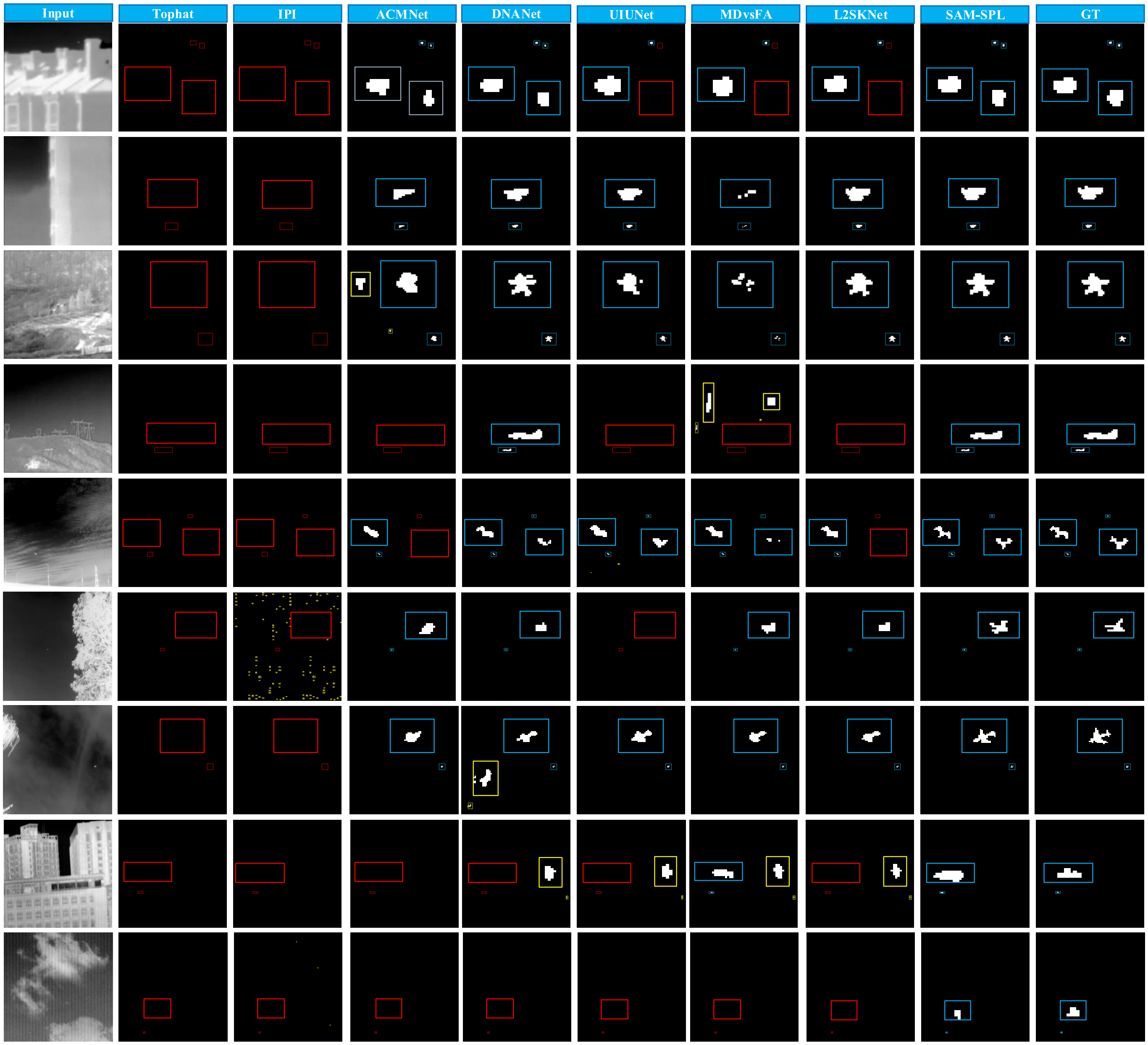


Fig. 5. Visualization of predictions produced by different methods on the NUDT-SIRST, IRSTD-1K, and IRSTDID-SKY datasets. The correct detection, false alarm, and miss detection are highlighted by blue, yellow, and red rectangles, respectively. Enlarged regions of the detected targets are shown in the corner for clear illustration.

convolutional residual blocks as the detection backbone. The model is optimized using the adaptive Nesterov momentum algorithm (Adan) [72] for 300 epochs, with a batch size of 12 and an initial learning rate of 0.01. The first three stages of the SAM2 (Hiera-Tiny version) encoder are employed to provide semantic guidance. The experiments are conducted in the PyTorch framework with Nvidia RTX 4090 GPU support.

B. Performance Comparison

1) Comparison Methods: To validate the effectiveness of the proposed method in detecting various types of infrared small targets across different sensing platforms, we compare SAM-SPL with a wide range of state-of-the-art ISTD methods, including twelve model-driven methods (Tophat [8], Max-Median [9], LIG [10], MPCM [11], RLCM [12], TLLCM [13], WSLCM [14], MSLSTIPT [15], IPI [16], RIPT [17], NRAM [32], and PSTNN [33]) and thirteen data-driven methods (ACMNet [34], ALCNet [35], ISNet [36], ResUNet [24], MDvsFA [25], DNANet [37], AGPCNet [38], UIUNet [40], MSHNet [41], P-MSHNet [42], RPCANet [43], L2SKNet [44], and MTUNet [47]).

2) Quantitative Results: The quantitative results of the proposed method and comparison methods on different datasets are presented in Tables I and II. According to the results, the performance of SAM-SPL demonstrates clear superiority over comparison methods in both pixel-level and target-level evaluations. First, on aerial- and land-based ISTD datasets

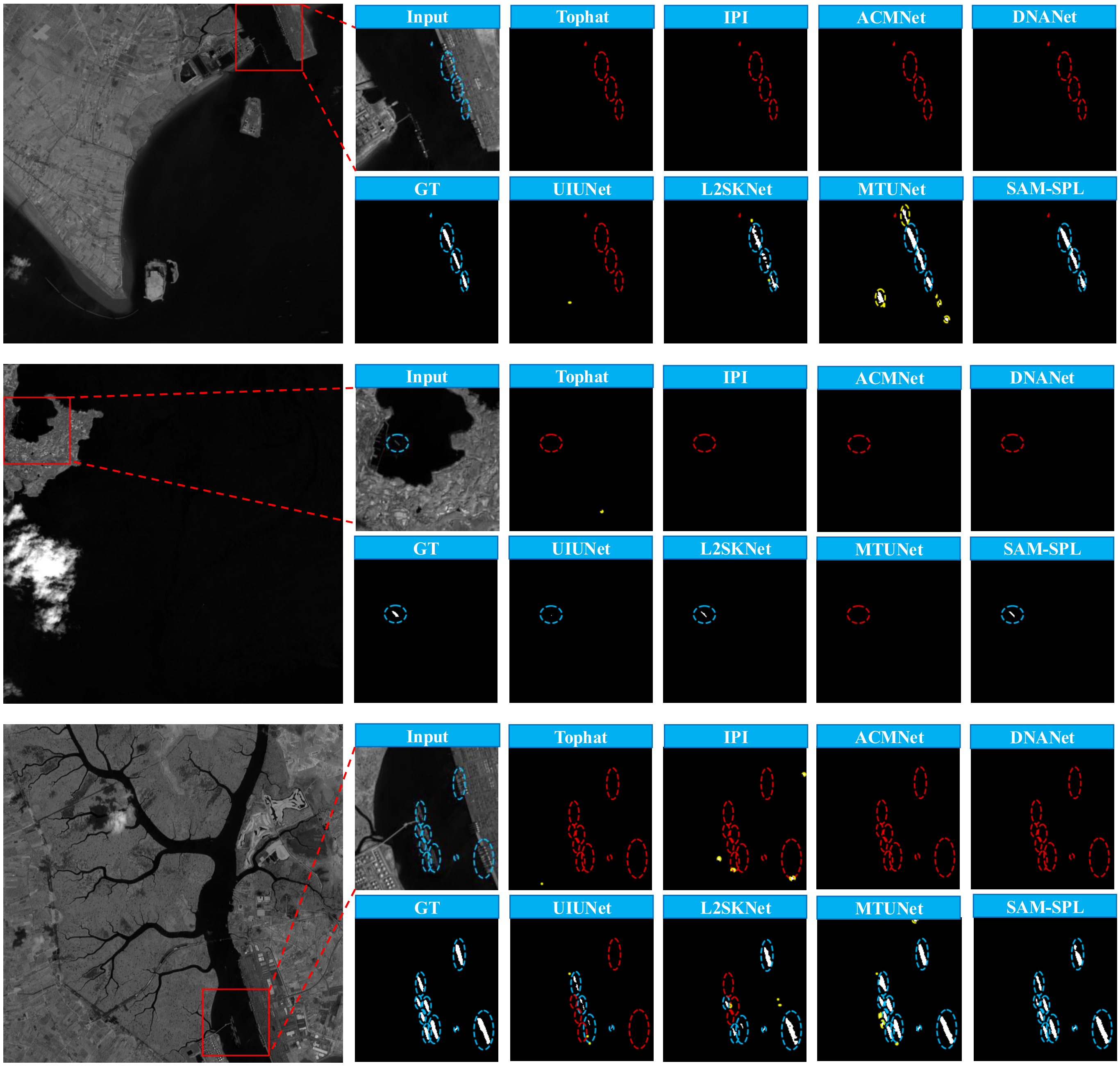


Fig. 6. Visualization of predictions produced by different ISTD methods on the NUDT-SIRST-Sea datasets. The correct detection, false alarm, and miss detection are highlighted by blue, yellow, and red circles, respectively. For clear illustration, the target-containing regions in the input images, ground truths, and model predictions are enlarged.

with relatively short imaging distances, SAM-SPL achieves the highest IOU and F_1, with overall improvements of about 4% and 3%, respectively. Then, target-level evaluations show that the proposed method achieves more favorable tradeoffs between miss detection and false alarms, as it increases P_d while simultaneously reducing F_a. In contrast, the comparison methods struggle to excel in both P_d and F_a, failing to ensure detection accuracy and reliability simultaneously. Although the proposed method yields a slightly higher F_a than UIUNet on the NUDT-SIRST dataset, it achieves a significantly higher P_d, indicating superior overall effectiveness. Notably, the proposed method achieves near-perfect performance on the IRSTDID-SKY dataset, successfully detecting almost all targets with an exceptionally low false alarm rate. Moreover, due to smaller sizes and greater scale variation, detecting tiny ships from satellite-based infrared images is more challenging. As reported in Table II, SAM-SPL still achieves the best performance, even compared to specialized architectures tailored for this task scenario, that is, MTUNet. In addition to the fixed-threshold evaluations, we also plot the ROC curves of different methods. The corresponding curves of each method are displayed in Fig. 4. We can see that the ROC curves of SAM-SPL consistently enclose the largest area across different datasets. As the false-alarm rate increases, the curve of

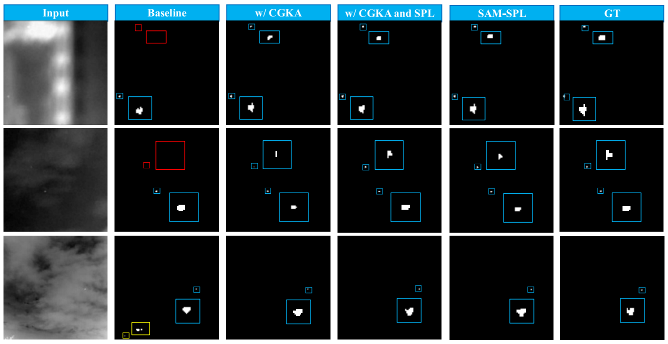


Fig. 7. Visual comparison of detection results with key component variants. The correct detection, false alarm, and miss detection are highlighted by blue, yellow, and red rectangles, respectively.

TABLE III

Ablation Experiments to Prove the Effectiveness of Each Component of the Proposed Method. Best Results Are Highlighted in Bold

Component	CGKA	SPL	MSIC	IoU	F_1	P_d	F_a
Baseline				65.82	79.38	90.91	25.51
w/ CGKA	✓			68.27	81.15	89.56	14.63
w/ SPL		✓		66.03	79.54	88.55	25.83
w/ MSIC			✓	65.93	79.46	91.58	15.56
w/o MSIC	✓	✓		71.40	83.31	90.57	11.98
w/o SPL	✓		✓	72.72	84.20	91.58	11.29
w/o CGKA		✓	✓	72.48	84.04	91.25	10.48
SAM-SPL	✓	✓	✓	**74.09**	**85.11**	**92.59**	**9.28**

SAM-SPL rises rapidly and remains superior to those of other methods. This further proves that the proposed method can accurately perceive small targets in infrared images while effectively controlling false alarms.

3) Qualitative Results: Apart from the quantitative results above, we further compare the qualitative results of our proposed method with several representative ISTD methods. Visualization examples on different datasets are provided in Figs. 5 and 6. First, conventional model-driven methods exhibit a large gap compared to data-driven methods, often producing substantial missed detections and false alarms in their predictions. Then, as shown in Fig. 5, for targets in infrared images captured by aerial-based platforms, the proposed method can accurately localize visually indistinct small targets from complex backgrounds while avoiding false alarms. In addition, compared to other methods, SAM-SPL can more precisely delineate the morphology of the detected targets with various shapes. This advantage further validates the effectiveness of our framework in reconciling detail preservation and contextual understanding. Even in the presence of external noise perturbations (as illustrated in the last two rows), SAM-SPL can still produce reliable detection outcomes. Furthermore, according to the results shown in Fig. 6 for satellite-based infrared ship detection under long imaging distances, SAM-SPL remains effective in accurately localizing targets across wide fields of view, while providing fine-grained descriptions of geometrical characteristics. For instance, as shown in the last row of Fig. 6, most methods fail to detect small ship targets that are densely arranged in the port. Although the MTUNet succeeds in detecting the targets, it also introduces false alarms by incorrectly focusing on background noise. Besides, the shape segmentation results of other methods cannot reflect the underlying geometrical characteristics of ship targets with varying sizes. In comparison, the proposed method can formulate robust awareness for various target types across different infrared sensing platforms under the semantic guidance of the SAM.

C. Ablation Studies

1) Contribution of Each Component: The proposed unified ISTD framework comprises three main components: the CGKA, the SPL, and the mutual semantic inconsistency calibration (MSIC). In this section, we conduct ablation experiments on the IRSTD-1K dataset to prove the contribution of each component.

Both pixel- and target-level performances are reported in Table III. We adopt the naive U-Net as the baseline, which is trained by the multiscale supervision strategy. First, incorporating the semantic guidance of SAM through CGKA leads to significant improvements in the overall detection performance, particularly in pixel-level evaluations. This validates the efficacy of our proposed learning pipeline in facilitating task-specific contextual understanding. Then, we observe that the individual use of SPL and mutual calibration yields certain improvements, and combining them with knowledge adaptation can further enhance the detection performance. This demonstrates their respective effectiveness in complementing target details and mitigating semantic inconsistency. However, even when used in conjunction, the resulting performance remains unsatisfactory without SAM guidance, due to the semantic-detail bottleneck inherent in learning from scratch. In addition to the quantitative analysis, we further conduct a qualitative analysis to better illustrate the contribution of each component to the detection of infrared targets. The detection results of incrementally incorporating each component into the baseline are visualized in Fig. 7. According to the results, the miss detections and false alarms can be effectively alleviated by incorporating CGKA, while SPL complements low-level details to enable more precise morphology capture. Finally, combining all three components into SAM-SPL leads to the best ISTD performance, which is evidenced by both quantitative results and qualitative visualizations.

2) Knowledge Adaptation Manner: During the encoding stage, we perform knowledge adaptation in a consult-guide manner, which is the core for enhancing contextual understanding. To verify the rationality of our knowledge adaptation strategy, we investigate the impact of providing semantic guidance through different strategies on the NUDT-SIRST and IRSTDID-SKY datasets.

Specifically, the illustrations of different knowledge adaptation strategies are shown in Fig. 8. The subfigures correspond to: (a) directly using the inference of SAM as semantic guidance; (b) using the integrated representation as the input of SAM blocks; and (c) generating semantic queries solely based on representations from convolutional residual blocks.

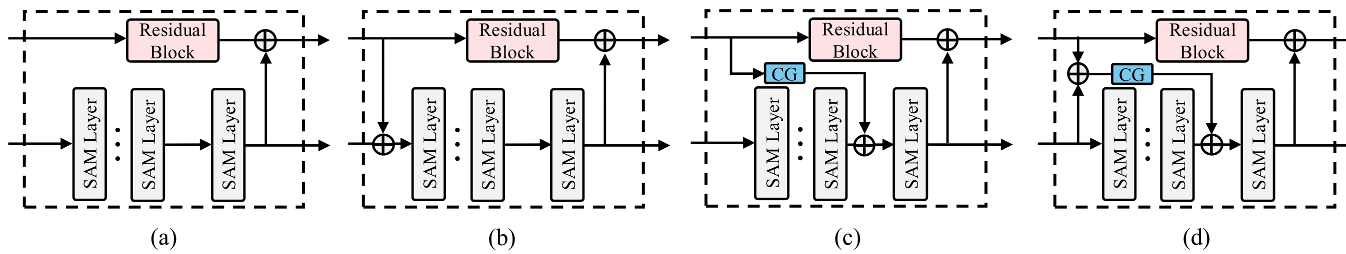


Fig. 8. Illustrations of different knowledge adaptation strategies to leverage the general knowledge encoded in the SAM.

TABLE IV

ABLATION EXPERIMENTS FOR DIFFERENT KNOWLEDGE ADAPTATION STRATEGIES. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Strategy	NUDT-SIRST			IRSTDID-SKY		
	IoU	P_d	F_a	IoU	P_d	F_a
None	91.38	98.31	6.85	65.22	96.15	2.80
(a)	90.64	98.20	9.86	63.79	96.79	6.34
(b)	90.91	98.41	9.10	66.35	96.15	1.40
(c)	90.15	98.10	7.17	65.78	94.87	3.08
(d)	**94.63**	**99.47**	**2.55**	**73.40**	**98.72**	**0.97**

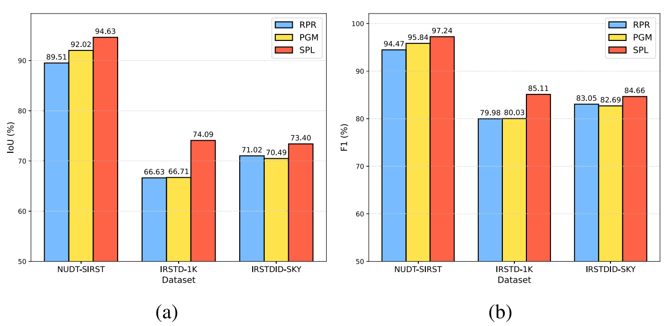


Fig. 9. Ablation experiments for different prompt learning strategies. (a) IoU. (b) F_1.

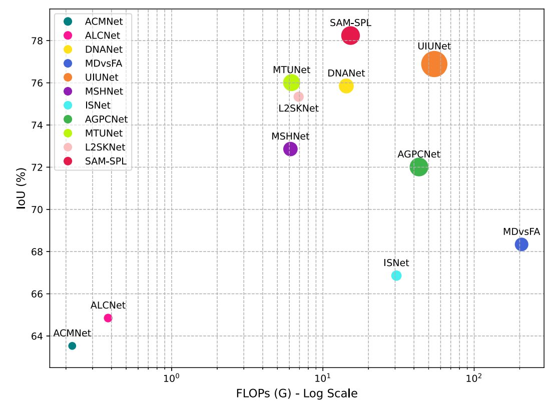


Fig. 10. Computational efficiency analysis of different methods.

The detection performance is reported by IoU, P_d, and F_a in Table IV. "None" represents learning without incorporating the inference of SAM. According to the results, directly using the inference of SAM as semantic guidance not only fails to enhance performance but leads to degradation instead, primarily due to the lack of interaction between the SAM and convolutional blocks. Then, integrating representations from convolutional residual blocks into the SAM inference process yields some performance improvements, but remains marginal. Moreover, generating semantic inquiries solely from the representations of convolutional blocks causes overall performance degradation, since the effectiveness of semantic guidance is severely limited without the consult process. In comparison, formulating semantic inquiries from the integrated representations achieves the best detection performance.

3) Prompt Learning Strategy: In addition to enhancing the contextual understanding by CGKA, we also employ self-derived prompts to complement low-level details in the deep semantic space. To demonstrate the effectiveness of our designed SPL, we compare the detection performance of the proposed ISTD framework when combined with different prompting strategies. Specifically, we compare SPL against two alternatives, where prompt tokens are generated by the relative position representations (RPRs) [73] and the input-conditioned prompt generation module (PGM) [67], respectively. The IoU and F_1 on different datasets are shown in Fig. 9. We can see that the proposed SPL consistently leads to the best detection performance.

4) Computational Efficiency: In this experiment, we analyze the computational efficiency of the proposed method by comparing its computational complexity and overall detection performance with other methods. Specifically, the computational complexity is measured by two most commonly adopted metrics in ISTD tasks, that is, the number of parameters and floating-point operations (FLOPs). The models are jointly trained on three aerial- and land-based ISTD datasets (NUDT-SIRST, IRSTD-1K, and IRSTDID-SKY) with the input image size of 256 $\times$ 256. As shown in Fig. 10, the overall detection performance is reported by the IoU across different datasets, while the area of each circle corresponds to the number of parameters of each method. We can see that SAM-SPL achieves superior overall performance without incurring excessive computational overhead, demonstrating high computational efficiency.

5) Performance When Meeting Variabilities: In real-world applications, the presence of variabilities in the collected infrared images is inevitable, which poses additional challenges to the perception of targets. In this experiment, we evaluate the performance of the proposed method on the IRSTD-1K dataset when meeting various variabilities. First,

TABLE V
DETECTION PERFORMANCE WITH VARIOUS VARIABILITIES. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Image Size			Deviation Threshold		
	128	256	512	1	2	3
ACMNet	19.8	43.1	62.9	77.1/64.2	87.2/18.2	90.6/17.3
ISNet	29.5	56.7	64.7	78.8/89.3	86.9/50.1	87.2/15.0
L2SKNet	41.1	52.2	65.6	83.8/52.5	90.9/20.2	92.3/10.8
SAM-SPL	**42.7**	**57.3**	**74.1**	**84.2/44.3**	**90.9/14.5**	**92.6/9.3**

TABLE VI
DETECTION PERFORMANCE WITH DIFFERENT PARAMETER SETTINGS. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Parameter	Setting	Params	IoU	F_1	P_d	F_a
Depth	4	6.04M	93.3	96.5	99.2	2.8
	5	12.25M	**94.6**	**97.2**	**99.5**	**2.6**
	6	36.76M	91.5	95.6	98.6	6.3
Layers	[1]	11.98M	90.2	94.8	98.1	11.1
	[1, 2]	12.05M	89.7	94.6	97.5	8.2
	[1, 2, 3]	12.25M	**94.6**	**97.2**	**99.5**	**2.6**
	[1, 2, 3, 4]	12.63M	92.2	95.9	98.9	2.7

we compare the IoU of different methods when changing the input image resolution from 128 to 512. According to the results shown in Table V, the proposed method consistently outperforms other methods across various image resolutions. Then, we also report the target-level performance (i.e., P_d/F_a) with different settings of deviation threshold. The results demonstrate that SAM-SPL can steadily achieve more accurate target detection while effectively suppressing false alarms.

D. Parameter Analysis

In this section, we analyze the impact of parameter settings in the proposed unified ISTD framework, including the number of convolutional blocks in the encoder (i.e., encoder depth) and the selection of shallow-layer feature maps for prompt generation. The experiments are carried out on the NUDT-SIRST dataset.

First, we evaluate the detection performance of the proposed method with different encoder depths. Specifically, we vary the number of convolutional blocks from 4 to 6, corresponding to 2 to 4 SAM blocks for semantic guidance. According to the results shown in Table VI, a shallow encoder fails to capture holistic contextual semantics due to insufficient knowledge adaptation, resulting in unsatisfactory detection performance. Conversely, an excessive increase in encoder depth introduces extra parameters without further performance gains, but it even leads to performance degradation. This is because the gain from enhanced contextual understanding is outweighed by the loss of target details. Therefore, a moderate encoder depth is more appropriate. Then, based on the determined encoder depth of 5, we compare the performance when generating prompt tokens from different layers. According to the results, incorporating richer shallow-layer features can better complement low-level target details within the deep semantic space. However, introducing features from deeper layers yields adverse effects, since redundant abstract semantics tend to obscure spatial textures. Based on the above analysis, we set the encoder depth to 5 and use the output feature maps from the first three residual blocks to generate self-derived prompts.

V. CONCLUSION

In this article, we have proposed SAM-SPL, a unified framework for ISTD. Instead of learning from scratch, SAM-SPL adapts general knowledge from the SAM to facilitate task-specific contextual understanding. Specifically, the inference of the SAM is incorporated with encoder blocks through a consult-guide manner to provide semantic guidance. Then, self-derived prompts are generated from shallow-layer features, which bidirectionally interact with latent representations to complement low-level target details. Subsequently, a mutual calibration module is integrated into the skip connection to mitigate semantic inconsistency, ensuring coherent spatial–semantic fusion during resolution recovery. Moreover, the model is optimized by a multiscale supervision strategy to formulate robust awareness for targets with varying sizes and shapes. Extensive evaluations have been conducted on various ISTD datasets, and SAM-SPL consistently achieves superior performance across diverse platforms.

Nevertheless, the proposed method still has certain limitations in complex real-world scenarios. In practical applications, sensing conditions are often dynamic and unpredictable, leading to inevitable distribution shifts and external perturbations during target perception. While SAM-SPL maintains advantages over existing methods when meeting different variations, it also suffers from nonnegligible performance degradation. In addition, its computational complexity may fail to satisfy real-time requirements in resource-constrained conditions. Therefore, our future exploration will focus on enhancing the reliability and efficiency of ISTD in practical deployments, particularly in terms of cross-domain generalizability, adversarial robustness, and lightweight design.

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